ALWYN ROY

DevOps Engineer

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3+ years building and operating cloud infrastructure on AWS — from zero-touch CI/CD pipelines and Terraform-managed environments to a GitOps observability platform running Prometheus, Loki, Tempo, and Grafana on Kubernetes. Cut monthly AWS spend by 15% through targeted rightsizing and RI/Savings Plan optimisation. Reduced deployment effort by 70% by eliminating all manual handoffs across five containerised services.

PROFESSIONAL EXPERIENCE

DevOps Engineer (Cloud & FinOps) | **Abilytics Consulting Pvt Ltd** *Aug 2022 – Present · Kochi, India*

- Designed and shipped a source-to-production CI/CD pipeline (GitHub Actions + Jenkins → Docker → ECR → ECS) covering five containerised services. Standardised image tagging and ECS deployment strategies; deployment effort dropped 70% and environment drift was eliminated.
- Replaced 100% of manually provisioned AWS infrastructure with modular Terraform — EC2, ECS, RDS, S3, VPC, IAM, and Security Groups. Environment provisioning dropped from hours to under 10 minutes; rollbacks became a single command across Dev, Staging, and Production.
- Reduced monthly AWS spend by 15% without breaching an SLA: rightsized EC2 instances, purged unattached EBS volumes and idle resources, enforced mandatory tagging, and converted on-demand spend to Reserved Instances and Savings Plans.
- Deployed CloudHealth and Cloudability across a multi-account AWS Organisation to expose per-team and per-environment spend. Replaced ad-hoc finance queries with executive dashboards surfacing cost waste that had previously been invisible.
- Built a serverless anomaly detection pipeline (Lambda + EventBridge + SNS) that scans for budget overruns daily and fires automated alerts to engineering and management — reclaiming ~2 engineering hours per week previously spent on manual finance reviews.
- Cut Mean Time to Detect (MTTD) for production incidents by 40% by building centralised CloudWatch dashboards and alarms across ECS, API Gateway, RDS, and Lambda.
- Enforced least-privilege IAM policies, encryption at rest and in transit, mandatory resource tagging, and CloudTrail audit logging — cleared all internal security and compliance reviews with zero critical findings.
- Automated routine Linux instance operations (Ubuntu and Amazon Linux) using Bash and Python, eliminating a class of recurring manual tasks and reducing operational toil.

KEY PROJECTS

GitOps-Managed Full-Stack Observability Platform

ArgoCD · Kubernetes · Next.js · NestJS · MySQL · Prometheus · Grafana · Loki · Tempo · Alloy · OpenTelemetry · Faro

- Deployed a Next.js frontend and NestJS backend on Kubernetes (Minikube/EC2) with MySQL, delivering all changes through ArgoCD — every Kubernetes manifest stored in Git and automatically synced to the cluster, removing all manual kubectl operations from the workflow.
- Configured ArgoCD with selfHeal and prune; validated drift detection and recovery through intentional drift injection tests. Cluster state is reconciled within 3 minutes across dev and production namespaces.
- Deployed a three-signal observability stack — metrics (Prometheus + Grafana), logs (Loki), and traces (Tempo + OpenTelemetry) — unified under Grafana Alloy as a single OTel-compatible collector, replacing multiple legacy agents.
- Instrumented the NestJS backend with the OpenTelemetry SDK for end-to-end distributed tracing; added Grafana Faro to the Next.js frontend for Real User Monitoring (LCP, FID, CLS, JS error tracking) — providing full frontend-to-database request visibility in Grafana.
- Defined SLOs (99.9% availability, p95 latency < 500 ms) with automated alerting; exposed custom business metrics (orders, revenue, active users) as Prometheus counters correlated against infrastructure health.

- Added mysqld_exporter as a sidecar for database-layer metrics; configured a synthetic traffic generator with 5% error injection and 2% artificial latency to validate alerting behaviour under realistic load.

End-to-End CI/CD Pipeline — Five Microservices

Jenkins · GitHub Actions · Docker · Amazon ECR · Amazon ECS

- Automated source-to-production delivery for five containerised microservices with zero manual handoffs and no configuration drift across environments. Deployment effort reduced by 70%.

Terraform IaC Migration

Terraform · EC2 · ECS · RDS · S3 · VPC · IAM · DynamoDB

- Converted 100% of manually managed AWS resources into reusable, parameterised Terraform modules with remote state (S3 + DynamoDB locking). Provisioning dropped from hours to under 10 minutes; infrastructure rollbacks became single-command operations.

FinOps Cost Optimisation Programme

CloudHealth · Cloudability · AWS Cost Explorer · CUR · Cost Anomaly Detection

- Delivered a sustained 15% reduction in monthly AWS spend through rightsizing, automated zombie-resource cleanup, tagging enforcement, and RI/Savings Plan optimisation. Self-serve executive dashboards replaced manual finance reporting.

Serverless Cost Monitoring & Alerting

AWS Lambda · EventBridge · SNS · CloudWatch · Python

- Built a zero-maintenance event-driven alerting system that detects budget anomalies daily. Cost-spike response time dropped from days to hours; reclaimed ~2 engineering hours per week.

MERN Stack Deployment & Infrastructure Automation

MERN · Docker · Nginx · EC2 · S3 · GitHub Actions

- Automated the full deployment of a MERN application on EC2 — containerised with Docker, served behind Nginx, with static assets hosted on S3 and delivery orchestrated via GitHub Actions.

TECHNICAL SKILLS

GitOps & CD	ArgoCD (App of Apps, ApplicationSets, selfHeal, drift correction), Kustomize, Helm
AWS	EC2, ECS, EKS, RDS, S3, VPC, IAM, Lambda, ECR, CloudWatch, Cost Explorer, Organizations, CloudTrail, SNS, EventBridge, API Gateway
IaC	Terraform (modules, remote state, S3 backend, DynamoDB locking), AWS CloudFormation
CI/CD	Jenkins, GitHub Actions, Docker, Amazon ECR, ECS (Rolling/Blue-Green), Bash, Python
Kubernetes	Minikube, Helm, Kustomize, ConfigMaps, Deployments, ServiceMonitors, ArgoCD, NodePort, RBAC
Observability	Prometheus, Grafana, Loki, Tempo, OpenTelemetry, Grafana Alloy, Grafana Faro (RUM), PromQL, CloudWatch
FinOps	CloudHealth, Cloudability, AWS Cost Explorer, CUR, Cost Anomaly Detection, Reserved Instances, Savings Plans
Security	IAM Least-Privilege, Security Groups, VPC Networking, Encryption, CloudTrail, AWS Config
OS & Scripting	Linux (Ubuntu, Amazon Linux), Bash, Python, Shell Automation, systemd
Source Control	Git, GitHub, GitOps, Agile/Scrum
Networking	VPC, Subnets, Route Tables, NAT Gateway, Load Balancers, Route 53, iptables

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (CSA-A) · Amazon Web Services

EDUCATION

M.Sc. Computer Science *2020 – 2022*
St. Berchmans College (Autonomous), Changanassery

B.Sc. Computer Science *2017 – 2020*
College of Applied Science, Kattappana